

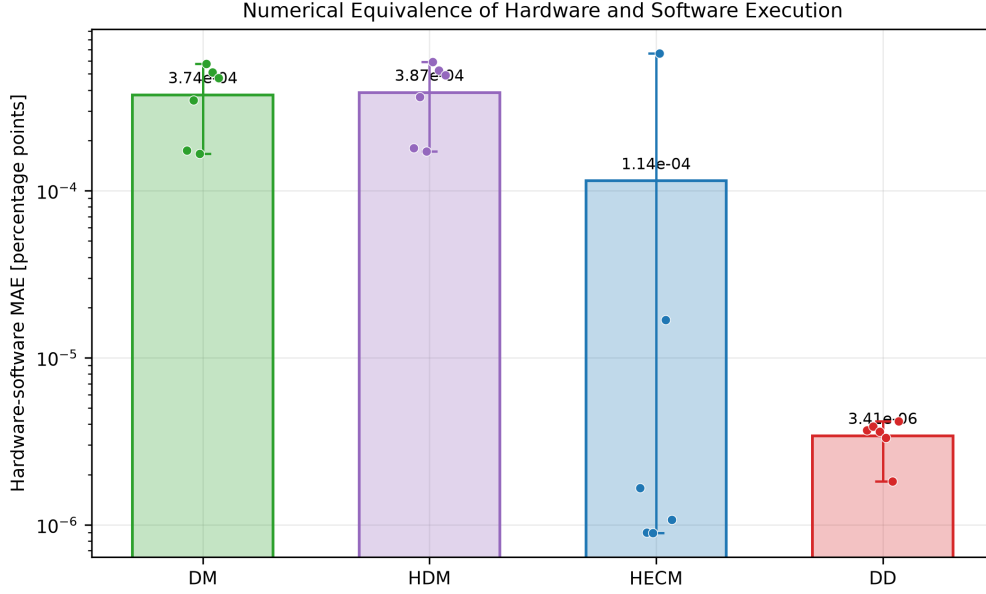


Figure 6.17: Numerical equivalence of STM32 and software execution over six cell sequences. Bars show mean hardware-to-software MAE in percentage points, dots show cells, and whiskers show the cell range. The logarithmic scale emphasizes that implementation differences are negligible relative to estimator error.

DM and HDM have sub-microsecond median latency at 0.171 and $0.165 \mu\text{s}$. HECM requires $23.9 \mu\text{s}$, while continuous-state DD requires $424.9 \mu\text{s}$. Figure 6.18 shows the complete distributions. The ordering follows the operations performed at each update. DM and HDM require a small number of scalar calculations. HECM additionally performs parameter lookup and observer matrix operations. DD evaluates recurrent matrix products and its MLP output head. At the nominal 1-Hz update rate, even the largest median occupies less than 0.043% of one sampling period. This result applies to the isolated estimator task and quantifies the execution margin available for other BMS functions.